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### HIGHER ACCURACY SECURE PHASE-BASED RANGING AND DIRECTION FINDING

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#### Abstract

A system and method for improving the accuracy of a secure phase-based ranging procedure and a Direction Finding procedure. The method includes receiving radio frequency signals from a second communication device. The method includes operating in a first mode including generating first location data based on the radio frequency signals and, transferring the first location data to a second processor in compliance with a Bluetooth Host Control Interface. The method includes comparing one or more conditions to one or more threshold values and responsive to the comparing transitioning from operating in the first mode to operating in a second mode. The method includes, while operating the second mode, generating second location data based on the radio frequency signals and, transferring the second location data to the second processor at a higher data transfer rate than the transferring of the first location data to the second processor.

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## **Background/Summary**

**RELATED APPLICATIONS** [0001] The present application is a continuation application of U.S. Non-Provisional application Ser. No. 17/527,104 filed on Nov. 15, 2021, which is incorporated by reference herein in their entirety.

### **TECHNICAL FIELD**

[0002] The present disclosure relates generally to the field of wireless telecommunications, and more particularly, to enhanced systems and methods for improving the accuracy of wireless device location measurements.

### **BACKGROUND**

[0003] Narrow-band radios such as Bluetooth Low Energy (LE) or IEEE 802.15.4 radios may determine the location of a wireless device relative to another wireless device. Procedures for determining the location (e.g., distance, angle) of wireless devices may include secure phase-based ranging procedures and the Direction Finding (DF) procedures. These procedures may apply a phase-based approach or a round-trip time (RTT) approach.

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## **Description**

### **BRIEF DESCRIPTION OF THE DRAWINGS**

[0004] The present disclosure is illustrated by way of example, and not by way of limitation, in the figures of the accompanying drawings in which like reference numerals refer to similar elements and in which:

[0005] **FIG. 1** illustrates a block diagram of an example Bluetooth system that improves the accuracy of wireless device location measurements, according to some embodiments;

[0006] **FIG. 2** is a flow diagram of a method for selecting between a standard mode and an enhanced proprietary mode, according to some embodiments;

[0007] **FIG. 3** illustrates a block diagram of an example mode selector for selecting between a standard mode and an enhanced proprietary mode, according to some embodiments; and

[0008] **FIG. 4** is a flow diagram of a method for determining the location of wireless devices using different modes of operation.

### **DETAILED DESCRIPTION**

[0009] The following description sets forth numerous specific details such as examples of specific systems, components, methods, and so forth, in order to provide a good understanding of various embodiments of the techniques described herein for improving the accuracy of location measurements, for example using secure phase-based ranging procedures and Direction Finding (DF) procedures for determining the location of wireless devices. It will be apparent to one skilled in the art, however, that at least some embodiments may be practiced without these specific details. In other instances, well-known components, elements, or methods are not described in detail or are presented in a simple block diagram format in order to avoid unnecessarily obscuring the techniques described herein. Thus, the specific details set forth hereinafter are merely exemplary.

Particular implementations may vary from these exemplary details and still be contemplated to be within the scope of the present disclosure.

[0010] Wireless technologies such as Bluetooth (BT) use positioning technologies, such as secure phase-based ranging and Direction Finding (DF) to determine the position/location of wireless devices. In secure phase-based ranging, a distance measurement application may determine the distance between two wireless devices based on a phase-based approach or a round-trip time (RTT) approach. In the phase-based approach, the two wireless devices exchange Constant Tones (CT) signals and one device performs a phase and/or frequency measurement based on the CT signals. In the RTT approach, the two wireless devices exchange RTT packets and timing reports/measurements.

[0011] Direction Finding may be based on the concepts of angle of arrival (AoA) and angle of departure (AoD). Direction finding can make use of the angular phase-shifts that occur between the antennas of a wireless device that receives (AoA) or transmits (AoD) radio frequency (RF) signals. With the use of antenna arrays at either side of the communication link, phase shift data can be determined and from this, location and angle of direction can be calculated.

[0012] Both the secure phase-based ranging and Direction Finding procedures may rely, at least in some part, on a digitized radio signal represented in an in-phase and quadrature-phase (I/Q) format, from which distance information and angular information is extracted. In embodiments, a Bluetooth radio of a Bluetooth system (e.g., including a BT radio, a BT processor, and/or a memory, etc.) is capable of generating the digitized radio signal by sampling the incoming radio signal (sometimes referred to as, radio frequency signal) that it receives at its antenna using a high sampling rate (e.g., 12 megahertz (MHz)). The BT system may report secure phase-based ranging and Direction Finding results to an application executing on a host via a Bluetooth Host Control Interface (HCI), such as a Universal Asynchronous Receiver-Transmitter (UART).

[0013] For several reasons, however, the secure phase-based ranging and Direction Finding procedures may produce inaccurate results. First, the Bluetooth Location Services requires a Bluetooth system to perform the secure phase-based ranging and/or Direction Finding procedures on an incoming signal using a low sampling rate to generate a low-bandwidth result (sometimes referred to as, "low-bandwidth data"). According to the Bluetooth Location Services requirement, the low-bandwidth result of the secure phase-based ranging procedure does not include raw I/Q data/samples, but rather various phase and time measurements that are derived from the captured radio signal. The low-bandwidth result of the Direction Finding procedure is generated by sampling the raw I/Q dataset using a low sampling rate (e.g., 1 MHz), even though the Bluetooth system is capable of sampling at a high sampling rate (e.g., 12 MHz).

[0014] The Bluetooth Location Services prefer the low-bandwidth output of the secure phase-based ranging and/or Direction Finding procedures in order to meet transport requirements outside of the Bluetooth system. These requirements are considered in the Bluetooth Location Services as the Bluetooth HCI, which is a low-bandwidth interface. The Bluetooth system must reduce the bandwidth of the secure phase-based ranging and Direction Finding results in order for the Bluetooth system to be able to transfer the result, as a low-bandwidth result, across the low-bandwidth UART. In other words, the amount of data that can be delivered outside of the Bluetooth system is limited by the maximum speed (e.g., 3 MHz or 4 MHz) of its UART interface. Consequently, only a portion of the data that is generated by the Bluetooth system (e.g., by the BT modem) can be exported (e.g., transferred) outside of the Bluetooth system.

[0015] Second, the processor that is built into the Bluetooth system has less processing power than the processor that is built into the Micro Control Unit (MCUs) that is coupled to the Bluetooth system. As a result, the amount of calculation the Bluetooth system can perform to convert the raw I/Q dataset that is sampled at the high sampling rate (e.g., 12 MHz) to the lower-bandwidth format is limited, which leads to a non-optimal performance (e.g., low resolution, less accurate result) of the secure phase-based ranging and Direction-Finding procedures.

[0016] Third, because the algorithm that processes the raw I/Q dataset is within the Bluetooth system, it is therefore inaccessible to a customer's host device that is coupled to the Bluetooth system. Therefore, even if a customer has a preferred algorithm that is different than what is provided in the Bluetooth system's firmware, the customer cannot implement it because they do not have access to the raw I/Q dataset. The conventional systems implement precisely the data output format (e.g., low-bandwidth result) that is required in the Bluetooth Location Services, where there is no raw I/Q data available for the secure phase-based ranging procedure, and only a low sampling rate (e.g., 1 MHz) for the Direction Finding procedure; thereby preventing customer choice in algorithms used for these procedures.

[0017] Aspects of the disclosure address the above-noted and other deficiencies by disclosing enhanced systems and methods for implementing a secure phase-based ranging and Direction Finding procedure to accurately determine the position/location of wireless devices. As described in the below passages, the BT system utilizes two processors, each with their own memory (e.g., random access memory (RAM)) along with an additional, smaller shared memory pool that is accessible to both processors. The first processor is a low-performance processor that resides in a BT subsystem of the BT system and is configured to operate the components of the BT subsystem (e.g., BT radio, etc.). The second processor is a high-performance processor that resides in a Micro Control Unit (MCU) of the BT system.

[0018] The Bluetooth (BT) system is coupled to the MCU via two datapaths that allow the BT subsystem to report the results (e.g., including events) from the secure phase-based ranging and Direction Finding procedures to the MCU. The BT subsystem uses the first data path to report the low-bandwidth result of the secure phase-based ranging and Direction Finding procedures, as required by the Bluetooth Location Services. The BT subsystem uses the second data path to rapidly transfer large, raw I/Q data buffers to the higher performance processor (MCU). The MCU is programmable by a customer, such that it allows a customer to directly access (e.g., retrieve) the complete raw I/Q dataset in case the customer wishes to apply custom processing algorithms in the MCU.

[0019] The BT system includes a mode selector to control automated switching between the high-bandwidth mode and the low-bandwidth mode to consume less power when the higher performance is not necessary, as higher data throughput naturally consumes more power. Customers have a control mechanism to select the appropriate method of operation given various conditions and applications. Accordingly, embodiments described herein can be used to generate more accurate location determinations compared to conventional BT location procedures.

[0020] FIG. 1 illustrates a block diagram of an example Bluetooth system that improves the accuracy of wireless device location measurements. The BT system **100** (sometimes referred to as, "communication device") includes a Bluetooth (BT) subsystem **102** and a Micro Control Unit (MCU) subsystem **130**. In some embodiments, the BT subsystem **102** and the MCU subsystem **130** are on the same semiconductor chip. In some embodiments, the BT subsystem **102** and the MCU subsystem **130** are on different semiconductor chips, for example, the BT subsystem **102** is on a first semiconductor chip and the MCU subsystem **130** is on a second semiconductor chip.

[0021] The BT subsystem **102** includes an antenna **103** configured to receive one or more incoming signals from a wireless device (e.g., another BT system **100**, another BT subsystem **102**). The BT subsystem **102** includes a BT modem **104** configured to modulate signals using one or more modulation schemes (e.g., frequency-shift keying (FSK), phase-shift keying (PSK)) and transmit the modulated signals toward a wireless device (e.g., another BT system **100**, another BT subsystem **102**), via the antenna **103**.

[0022] In some embodiments, the BT modem **104** may be configured to receive one or more incoming signals from the wireless device responsive to transmitting the modulated signals toward the wireless device. In some embodiments, the BT modem **104** may be configured to generate I/Q data (sometimes referred to as, "raw I/Q dataset") based on the one or more incoming signals and

using a sampling rate (e.g., 1 MHz to 12 MHz). For example, the BT modem **104** may be configured to generate a first I/Q dataset using a 1 MHz sampling rate (sometimes referred to as, “low-sampling rate”) and generate a second I/Q dataset using a 12 MHz sampling rate (sometimes referred to as, “high-sampling rate”). In this instance, the second I/Q dataset contains more samples (e.g., data points) than the first I/Q dataset.

[0023] The BT system **100** includes a BT processing device **108**. The BT system **100** includes a BT memory **106** coupled to the BT processing device **108** configured to store instructions (including temporary data) that are executed by the BT processing device **108**.

[0024] The BT system **100** includes a shared memory **110** configured to store data (e.g., measurements, I/Q data, header and control data, etc.) to share with the MCU subsystem **130**. In some embodiments, the shared memory **110** includes an address spaces **112a**, **112b**. In some embodiments, an address space (e.g., a range of memory locations) may be any size, for example, 4 kilobytes (KB).

[0025] The BT system **100** includes a low-speed interface **123**, a high-speed interface **125**, and an inter-processor communication (IPC) interface **121**. One or more of the blocks and/or components of the BT subsystem **102** may be coupled together via an internal bus (not shown in FIG. 1) of the BT subsystem **102** to facilitate the transfer of data and/or messages between the components.

[0026] The MCU subsystem **130** includes an MCU processing device **138**. The MCU subsystem **130** includes an MCU memory (e.g., RAM) **136** coupled to the MCU processing device **138** for storing instructions (including temporary data) that are executed by the MCU processing device **138**. The MCU subsystem **130** includes a low-speed interface **123**, a high-speed interface **125**, and an IPC interface **131**. One or more of the blocks and/or components of the MCU subsystem **130** may be coupled together via an internal bus (not shown in FIG. 1) of the MCU subsystem **130** to facilitate the transfer of data and/or messages between the components.

[0027] In some embodiments, the low-speed interface **133** of the MCU subsystem **130** is coupled to the low-speed interface **123** of the BT subsystem **102** via a low-speed path **124**. In some embodiments, the high-speed interface **125** of the MCU subsystem **130** is coupled to the high-speed interface **125** of the BT subsystem **102** via a high-speed path **126**. In some embodiments, the IPC interface **131** of the MCU subsystem **130** is coupled to the IPC interface **121** of the BT subsystem **102** via an IPC path **122**.

[0028] In some embodiments, the low-speed interface **123** and the high-speed interface **125** of the BT subsystem **102** may be a single interface that supports both low-speed and high-speed communication. For example, the BT subsystem **102** use a single interface (e.g., low-speed interface **123** combined with high-speed interface **125**) to send a first dataset to the MCU subsystem **130** across the low-speed path **124** and a second dataset to the MCU subsystem **130** across the high-speed path **126**.

[0029] In some embodiments, the low-speed interface **133** and the high-speed interface **135** of the MCU subsystem **130** may be a single interface that supports both low-speed and high-speed communication. For example, the MCU subsystem **130** uses a single interface (e.g., low-speed interface **133** combined with high-speed interface **135**) to receive a first dataset from the BT subsystem **102** across the low-speed path **124** and a second dataset from the BT subsystem **102** across the high-speed path **126**.

[0030] In some embodiments, the MCU subsystem **130** uses the low-speed interface **133** to receive (e.g., transfer) a first dataset from the BT subsystem **102** across the low-speed path **124** at a first data transfer rate (e.g., maximum bit rate) and a second dataset from the BT subsystem **102** across the high-speed path **126** at a second data transfer rate, where the second data rate is higher than the first data rate.

[0031] In some embodiments, the BT modem **104** may be configured to generate an I/Q dataset (sometimes referred to as, “raw I/Q dataset”) based on the one or more incoming signals and using a sampling rate (e.g., 1 MHz to 12 MHz). For example, the BT modem **104** may be configured to

generate a first I/Q dataset using a 1 MHz sampling rate and generate a second I/Q dataset using a 12 MHz sampling rate. In this instance, the second I/Q dataset contains more samples (e.g., data points) than the first I/Q dataset. In some embodiments, an I/Q dataset includes data that is in an in-phase and quadrature-phase (I/Q) format.

[0032] In some embodiments, the BT subsystem **102** may be configured to perform a secure phase-based ranging procedure (sometimes referred to as, “location measurement”) to generate a secure phase-based ranging result (e.g., a phase measurement and/or timing reports/measurements that is shared with the MCU subsystem **130** via the low-speed path **124**, where the secure phase-based ranging result indicates a phase measurement and/or timing reports/measurements that combined with the other device's results determines the distance between the BT system **100** and the other wireless device. For example, the BT modem **104** may be configured to receive one or more incoming signals from another wireless device via antenna **103**. The BT modem **104** may be configured to generate I/Q data (sometimes referred to as, “high sample-rate (SR) I/Q data”) based on the one or more incoming signals using a high-sampling rate (e.g., 12 MHz). The BT modem **104** may be configured to perform a secure phase-based ranging procedure using the high-SR I/Q data (and possibly other data) and a first secure phase-based ranging algorithm to generate a secure phase-based ranging result (sometimes referred to as, “low-bandwidth secure phase-based ranging result”) that is compliant to the Bluetooth specification. That is, the BT modem **104** may be configured to use the first secure phase-based ranging algorithm to convert (e.g., format) the high-SR I/Q data into a low-bandwidth secure phase-based ranging result. The BT subsystem **102** may be configured to send (e.g., transfer, export) the low-bandwidth secure phase-based ranging result to the MCU subsystem **130** via the low-speed interface **123** and the low-speed path **124**, or the IPC interface **121** and the IPC path **122**.

[0033] In some embodiments, the BT subsystem **102** may be configured to perform a Direction Finding (DF) procedure (sometimes referred to as, “location measurement”) to generate a DF result that is shared with the MCU subsystem **130** via the low-speed path **124**, where the DF result indicates an angle (e.g., in degrees, radians) in which another wireless device is positioned with respect to the BT system **100**. For example, the BT modem **104** may be configured to receive one or more incoming signals from another wireless device via antenna **103**. The BT modem **104** may be configured to generate I/Q data (sometimes referred to as, “low-SR I/Q data”) based on the one or more incoming signals using a low sampling-rate (e.g., 1 MHz).

[0034] The BT modem **104** may be configured to send the low-SR I/Q data to the BT processing device **108** and/or store the low-SR I/Q data to the BT memory **106**. If the low-SR I/Q data is stored in the BT memory **106**, then the BT processing device **108** may fetch (e.g., retrieve) the low-SR I/Q data from the BT memory **106**. The BT processing device **108** may be configured to send the low-SR I/Q data to the MCU subsystem **130** via the low-speed interface **123** and the low-speed path **124**, or the IPC interface **121** and the IPC path **122**.

[0035] The MCU processing device **138** may be configured to perform, responsive to receiving the low-SR I/Q data, a DF procedure using the low-SR I/Q data and a first DF algorithm (sometimes referred to as, “a first customer-provided DF algorithm) to generate a DF result (sometimes referred to as, “low-bandwidth DF result”). The MCU processing device **138** may be configured to retrieve the first DF algorithm from the customer replaceable logic **137**.

[0036] In some embodiments, the BT subsystem **102** may be configured to perform a secure phase-based ranging procedure to generate a secure phase-based ranging result that is shared with the MCU subsystem **130** via the high-speed path **126**, where the secure phase-based ranging result indicates a phase measurement and/or timing reports/measurements that combined with the other device's results determines the distance between the BT system **100** and the other wireless device. For example, the BT modem **104** may be configured to receive one or more incoming signals from another wireless device via antenna **103**. The BT modem **104** may be configured to generate I/Q data (sometimes referred to as, “high sample-rate (SR) I/Q data”) based on the one or more

incoming signals using a high-sampling rate (e.g., 12 MHz). The BT modem **104** may be configured to send the high-SR I/Q data to the BT processing device **108**. The BT processing device **108** may be configured to store the high-SR I/Q data and the low-bandwidth secure phase-based ranging result (which is compliant to the BT specification, as discussed herein) to address space **112a** of the shared memory **110**. The BT processing device **108** may be configured to retrieve (e.g., obtain, transfer, acquire) the header and control data from the BT memory **106** and store the header and control data to the address space **112a** of the shared memory **110**.

[0037] The BT processing device **108** may be configured to send an IPC message to the MCU subsystem **130** via the IPC interface **121** and the IPC path **122**, where the IPC message indicates that data is available to be read from the address space **112a**. The MCU processing device **138** may be configured to fetch (e.g., obtain, read, acquire), responsive to receiving the IPC message, the high-SR I/Q data and the low-bandwidth secure phase-based ranging result from the address space **112a** via high-speed interface **135** and the high-speed path **126**. The MCU processing device **138** may be configured to perform a secure phase-based ranging procedure using the high-SR I/Q data and a second secure phase-based ranging algorithm (e.g., a customer-provided secure phase-based ranging algorithm) to generate a secure phase-based ranging result (sometimes referred to as, “high-bandwidth secure phase-based ranging result”).

[0038] In some embodiments, the first secure phase-based ranging algorithm and the second secure phase-based ranging algorithm are different. In some embodiments, the high-bandwidth secure phase-based ranging result is not compliant to the Bluetooth specification. In some embodiments, the high-bandwidth secure phase-based ranging result is more accurate than the low-bandwidth secure phase-based ranging result. For example, the low-bandwidth secure phase-based ranging result may indicate the phase measurement and/or timing reports/measurements that combined with the other device's results determines the distance between the BT system **100** and the other wireless device within a 100-150 nano-sec accuracy, and the high-bandwidth secure phase-based ranging result may indicate the phase measurement and/or timing reports/measurements that combined with the other device's results determines the distance between the BT system **100** and the other wireless device in less than a 15 nano-sec accuracy.

[0039] In some embodiments, the BT subsystem **102** may be configured to perform a Direction Finding (DF) procedure to generate a DF result that is shared with the MCU subsystem **130** via the high-speed path **126**, where the DF result indicates an angle (e.g., in degrees) in which another wireless device is positioned with respect to the BT system **100**. For example, the BT modem **104** may be configured to receive one or more incoming signals from another wireless device via antenna **103**. The BT modem **104** may be configured to generate I/Q data (sometimes referred to as, “high-SR I/Q data”) based on the one or more incoming signals using a high sampling-rate (e.g., 12 MHz). The BT modem **104** may be configured to send the high-SR I/Q data to the BT processing device **108**. The BT processing device **108** may be configured to store the high-SR I/Q data to address space **112a** of the shared memory **110**. The BT processing device **108** may be configured to retrieve the header and control data from the BT memory **106** and store the header and control data to the address space **112a** of the shared memory **110**.

[0040] The BT processing device **108** may be configured to send an IPC message to the MCU subsystem **130** via the IPC interface **121** and the IPC path **122**, where the IPC message indicates that data is available to be read from the address space **112a**. The MCU processing device **138** may be configured to fetch (e.g., obtain, read, acquire, retrieve), responsive to receiving the IPC message, the high-SR I/Q data from the address space **112a** via the high-speed interface **135** and the high-speed path **126**. The MCU processing device **138** may be configured to perform a DF procedure using the high-SR I/Q data and a second DF algorithm (e.g., a customer-provided DF algorithm) to generate a DF result (sometimes referred to as, “high-bandwidth DF result”). The MCU processing device **138** may be configured to retrieve the second DF algorithm from the customer replaceable logic **137**.

[0041] The BT processing device **108** may be configured to store the data/results (e.g., high-SR I/Q data and low-bandwidth secure phase-based ranging result) that were acquired from the secure phase-based ranging procedure for the high-speed path **126** to address spaces **112a**, **112b** of the shared memory **110** in a ping-pong fashion. For example, the BT processing device **108** may be configured to perform a secure phase-based ranging procedure using a first set of incoming signals to generate a first set of data/results (e.g., high-SR I/Q data and low-bandwidth secure phase-based ranging result), store the first set of data/results to the address space **112a** of the shared memory **110**, and send an IPC message to the MCU subsystem **130** via the IPC path **122**, where the IPC message indicates that data is available to be read from the address space **112a**. As the MCU subsystem **130** attempts to retrieve (or is retrieving) the first set of data/results from the address space **112a**, the BT processing device **108** may be configured to perform a secure phase-based ranging procedure using a second set of incoming signals to generate a second set of data/results (e.g., high-SR I/Q data and low-bandwidth secure phase-based ranging result), store the second set of data/results to the address space **112a** of the shared memory **110**, and send an IPC message to the MCU subsystem **130** via the IPC path **122**, where the IPC message indicates that data is available to be read from the address space **112b**. The BT processing device **108** may determine that the MCU subsystem **130** has retrieved the first set of data/results from address space **112a**, and proceed to repeat the secure phase-based ranging procedures using successive sets of incoming signals and storing the data/results in address spaces **112a**, **112b**. Likewise, the BT processing device **108** may be configured to store the data/results (e.g., high-SR I/Q data) that were acquired from the DF procedure for the high-speed path **126** to address spaces **112a**, **112b** of the shared memory **110** in a ping-pong fashion.

[0042] FIG. 2 is a flow diagram of a method for selecting between a standard mode and an enhanced proprietary mode, according to some embodiments. Although the operations are depicted in FIG. 2 as integral operations in a particular order for purposes of illustration, in other implementations, one or more operations, or portions thereof, are performed in a different order, or overlapping in time, in series or parallel, or are omitted, or one or more additional operations are added, or the method is changed in some combination of ways. In some embodiments, the method **200** may be performed by processing logic that comprises hardware (e.g., circuitry, dedicated logic, programmable logic, microcode, etc.), firmware, or a combination thereof. In some embodiments, some or all operations of method **200** may be performed by one or more components of the BT system **100** in FIG.

[0043] In some embodiments, the standard mode corresponds to the secure phase-based ranging procedure (as discussed herein) that generates a secure phase-based ranging result that is shared with the MCU subsystem **130** via the low-speed path **124**. In some embodiments, the standard mode corresponds to the DF procedure (as discussed herein) that generates a DF result that is shared with the MCU subsystem **130** via the low-speed path **124**.

[0044] In some embodiments, the enhanced proprietary mode corresponds to the secure phase-based ranging procedure (as discussed herein) that generates a secure phase-based ranging result that is shared with the MCU subsystem **130** via the high-speed path **126**. In some embodiments, the enhanced proprietary mode corresponds to the DF procedure (as discussed herein) that generates a DF result that is shared with the MCU subsystem **130** via the high-speed path **126**.

[0045] At operation **204**, in some embodiments, a mode selector **202** determines whether to run the standard mode or the enhanced proprietary mode. In some embodiments, the intelligent mode selector may make this determination based on one or more of a radio frequency (RF) quality, a local ToA-ToD quality, a phase measurement quality, a local agile multi-path or interference detection, a current angle in which the BT system **100** is with respect to another wireless device, or an override mode (e.g., as determined based on an override indicator and/or signal) that allows “locked” mode selection by the user. In some embodiments, the intelligent mode selector may be a component of the BT subsystem **102** or the MCU subsystem **130**.



[0046] If the standard mode is enabled, then the BT system **100** performs the secure phase-based ranging and/or DF procedures according to the standard mode by proceeding to operation **206**. At operation **206**, the BT system **100** uses its hardware to capture data. At operation **208**, the BT processing device **108** performs minimal processing on the capture data. For example, the BT processing device **108** samples/captures the data using a lower sampling rate (e.g., 1 MHz) than that which the BT processing device **108** is capable (e.g., 12 MHz). At operation **210**, the BT processing device **108** produces data/results that are complaint to the BT specification, where the data/results have a reduced accuracy as compared to that which is capable by the hardware (e.g., BT modem **104** in FIG. 1) of the BT system **100**.

[0047] If the enhanced proprietary mode is enabled, then the BT subsystem **102** performs the secure phase-based ranging and/or DF procedures according to the enhanced proprietary mode by proceeding to operation **212**. At operation **212**, the BT subsystem **102** uses its hardware to capture full data. For example, the BT processing device **108** samples/captures full data using a higher sampling rate (e.g., 12 MHz) than that which is required by the BT specification (e.g., 1 MHz). At operation **214**, the BT processing device **108** passes the full data to the MCU subsystem **130** via the IPC path **122**. At operation **216**, the MCU processing device **138** performs high-performance processing on the capture data. At operation **218**, the MCU processing device **138** produces a first set of data/results that are complaint to the BT specification and a second data/results that are not compliant to the BT specification, but are more accurate than the first set of data/results. At operation **220**, the MCU processing device **138** sends the first and second sets of data/results to a customer application/user interface (UI) executing on the MCU processing device **138** or executing on a separate device (e.g., a host device).

[0048] FIG. 3 illustrates a block diagram of an example intelligent mode selector for selecting between a standard mode and an enhanced proprietary mode, according to some embodiments. The intelligent mode selector **300** is configured to determine whether to run (e.g., execute, perform) the standard mode or the enhanced proprietary mode. In some embodiments, the intelligent mode selector may make this determination based on values and/or thresholds indicative of one or more of a radio frequency (RF) quality (e.g., Access Address Quality, RSSI, calculation quality), a current distance of the BT system **100** to another wireless device, a current angle in which the BT system **100** is with respect to another wireless device, a power budget (e.g., state of battery or battery power level), or an override mode that allows “locked” mode selection by the user. In some embodiments, the intelligent mode selector may be a component of the BT subsystem **102** or the MCU subsystem **130**.

[0049] In some embodiments, at larger distances the intelligent mode selector **300** may determine that a higher accuracy is not necessary, and therefore select a standard mode (e.g., a BT HCI standard mode, a coarse mode), but then run the enhanced proprietary mode when the distance becomes sufficiently small.

[0050] In some embodiments, once the intelligent mode selector detects that the level of RSSI is less than threshold, it chooses locally to switch from the standard mode to the enhanced proprietary mode, in an agile manner. In some embodiments, in case of secure phase-based ranging and Time of Arrival (ToA) estimation, an algorithm in the intelligent mode selector may decide to switch to the enhanced proprietary mode to post-process the raw IQ samples and to better estimate the ToA of the packet. In some embodiments, the decision of the intelligent mode selector may be chosen based on local-agile detection of interference (or multipath fading) on one particular channel.

[0051] FIG. 4 is a flow diagram of a method for determining the location of wireless devices include the secure phase-based ranging procedure and the Direction Finding procedure, according to some embodiments. Although the operations are depicted in FIG. 4 as integral operations in a particular order for purposes of illustration, in other implementations, one or more operations, or portions thereof, are performed in a different order, or overlapping in time, in series or parallel, or are omitted, or one or more additional operations are added, or the method is changed in some

combination of ways. In some embodiments, the method **400** may be performed by processing logic that comprises hardware (e.g., circuitry, dedicated logic, programmable logic, microcode, etc.), firmware, or a combination thereof. In some embodiments, some or all operations of method **400** may be performed by one or more components of the BT system **100** in FIG. **1**.

[0052] The method **400**, in some embodiments, includes the operation **402** of receiving radio frequency signals from a second communication device.

[0053] The method **400**, in some embodiments, includes the operation **404** of operating in a first mode including generating first location data based on the radio frequency signals and, using a first processor, transferring the first location data to a second processor in compliance with a Bluetooth Host Control Interface.

[0054] The method **400**, in some embodiments, includes the operation **406** of using the first processor, comparing one or more conditions to one or more threshold values and responsive to the comparing transitioning from operating in the first mode to operating in a second mode.

[0055] The method **400**, in some embodiments, includes the operation **408** of while operating the second mode, generating second location data based on the radio frequency signals and, using the first processor, transferring the second location data to the second processor at a higher data transfer rate than the transferring of the first location data to the second processor.

[0056] In the above description, some portions of the detailed description are presented in terms of algorithms and symbolic representations of operations on analog signals and/or digital signals or data bits within a non-transitory storage medium. These algorithmic descriptions and representations are the means used by those skilled in the data processing arts to most effectively convey the substance of their work to others skilled in the art. An algorithm is here and generally, conceived to be a self-consistent sequence of steps leading to a desired result. The steps are those requiring physical manipulations of physical quantities. Usually, though not necessarily, these quantities take the form of electrical or magnetic signals capable of being stored, transferred, combined, compared and otherwise manipulated. It has proven convenient at times, principally for reasons of common usage, to refer to these signals as bits, values, elements, symbols, characters, terms, numbers, or the like.

[0057] Reference in the description to “an embodiment,” “one embodiment,” “an example embodiment,” “some embodiments,” and “various embodiments” means that a particular feature, structure, step, operation, or characteristic described in connection with the embodiment(s) is included in at least one embodiment of the disclosure. Further, the appearances of the phrases “an embodiment,” “one embodiment,” “an example embodiment,” “some embodiments,” and “various embodiments” in various places in the description do not necessarily all refer to the same embodiment(s).

[0058] The description includes references to the accompanying drawings, which form a part of the detailed description. The drawings show illustrations in accordance with exemplary embodiments. These embodiments, which may also be referred to herein as “examples,” are described in enough detail to enable those skilled in the art to practice the embodiments of the claimed subject matter described herein. The embodiments may be combined, other embodiments may be utilized, or structural, logical, and electrical changes may be made without departing from the scope and spirit of the claimed subject matter. It should be understood that the embodiments described herein are not intended to limit the scope of the subject matter but rather to enable one skilled in the art to practice, make, and/or use the subject matter.

[0059] It should be borne in mind, however, that all of these and similar terms are to be associated with the appropriate physical quantities and are merely convenient labels applied to these quantities. Unless specifically stated otherwise as apparent from the above discussion, it is appreciated that throughout the description, discussions utilizing terms such as “receiving,” “generating,” “performing,” “sending,” “determining,” or the like, refer to the actions and processes of an integrated circuit (IC) controller, or similar electronic device, that manipulates and

transforms data represented as physical (e.g., electronic) quantities within the controller's registers and memories into other data similarly represented as physical quantities within the controller memories or registers or other such information non-transitory storage medium.

[0060] The words “example” or “exemplary” are used herein to mean serving as an example, instance, or illustration. Any aspect or design described herein as “example” or “exemplary” is not necessarily to be construed as preferred or advantageous over other aspects or designs. Rather, use of the words “example” or “exemplary” is intended to present concepts in a concrete fashion. As used in this application, the term “or” is intended to mean an inclusive “or” rather than an exclusive “or.” That is, unless specified otherwise, or clear from context, “X includes A or B” is intended to mean any of the natural inclusive permutations. That is, if X includes A; X includes B; or X includes both A and B, then “X includes A or B” is satisfied under any of the foregoing instances. In addition, the articles “a” and “an” as used in this application and the appended claims should generally be construed to mean “one or more” unless specified otherwise or clear from context to be directed to a singular form. Moreover, use of the term “an embodiment” or “one embodiment” or “an embodiment” or “one embodiment” throughout is not intended to mean the same embodiment or embodiment unless described as such.

[0061] Embodiments described herein may also relate to an apparatus (e.g., such as an AC-DC converter, and/or an ESD protection system/circuit) for performing the operations herein. This apparatus may be specially constructed for the required purposes, or it may comprise firmware or hardware logic selectively activated or reconfigured by the apparatus. Such firmware may be stored in a non-transitory computer-readable storage medium, such as, but not limited to, read-only memories (ROMs), random access memories (RAMs), EPROMs, EEPROMs, flash memory, or any type of media suitable for storing electronic instructions. The term “computer-readable storage medium” should be taken to include a single medium or multiple media that store one or more sets of instructions. The term “computer-readable medium” shall also be taken to include any medium that is capable of storing, encoding, or carrying a set of instructions for execution by the machine and that causes the machine to perform any one or more of the methodologies of the present embodiments. The term “computer-readable storage medium” shall accordingly be taken to include, but not be limited to, solid-state memories, optical media, magnetic media, any medium that is capable of storing a set of instructions for execution by the machine and that causes the machine to perform any one or more of the methodologies of the present embodiments.

[0062] The above description sets forth numerous specific details such as examples of specific systems, components, methods, and so forth, in order to provide a good understanding of several embodiments of the present disclosure. It is to be understood that the above description is intended to be illustrative and not restrictive. Many other embodiments will be apparent to those of skill in the art upon reading and understanding the above description. The scope of the disclosure should, therefore, be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled.

## Claims

**1-20.** (canceled)

**21.** A method of a first communication device, comprising: receiving radio frequency signals from a second communication device; sampling the radio frequency signals to generate digitized radio frequency signals; operating in a first mode including, using a first processor, formatting the digitized radio frequency signals into low-bandwidth results and transferring low-bandwidth results to a second processor in compliance with a Bluetooth Host Control Interface; based on a comparison of one or more conditions to one or more threshold values, using the first processor, transitioning from operating in the first mode to operating in a second mode; and while operating the second mode, using the first processor, transferring the digitized radio frequency signals to the

second processor at a higher data transfer rate than the transferring of the low-bandwidth results to the second processor.

**22.** The method of claim 21, wherein the comparison of the one or more conditions comprises, using the first processor, comparing one or more of a value indicative of a distance, a value indicative of a battery power level, or a value indicative of signal quality.

**23.** The method of claim 21, wherein the sampling the radio frequency signals to generate digitized radio frequency signals comprises using a sampling rate from 1 MHz to 12 MHz.

**24.** The method of claim 23, wherein the digitized radio frequency signals comprise I/Q data.

**25.** The method of claim 21, wherein the low-bandwidth results comprise distance or time data values.

**26.** The method of claim 1, further comprising transferring the low-bandwidth results over a first data path and transferring the digitized radio frequency signals over a second data path, wherein the first data path has a maximum data transfer rate that is lower than that of the second data path.

**27.** The method of claim 26, wherein the transferring of the digitized radio frequency signals to the second processor includes transferring the digitized radio frequency signals to a portion of a memory that is shared by the first process and the second processor.

**28.** The method of claim 27, wherein the transferring of the digitized radio frequency signals comprises, using the first processor, sending a message to the second processor to cause the second processor to retrieve the digitized radio frequency from the portion of the memory.

**29.** The method of claim 21, further comprising using the second processor, executing an algorithm using the digitized radio frequency signals to determine a distance associated with the second communication device.

**30.** A communication device, comprising: a radio frequency (RF) modem configured to: sample RF signals received from a second communication device to generate digitized RF signals; and a first processor configured to: operate in a first mode to transfer low-bandwidth results to a second processor in compliance with a Bluetooth Host Control Interface, wherein the low-bandwidth results are generated based on the digitized RF signals; compare one or more conditions to one or more threshold values; and transition, responsive to the comparing, from operating in the first mode to operating in a second mode; and while operating in the second mode, transfer the digitized RF signals to the second processor at a higher data transfer rate than the transferring of the low-bandwidth results to the second processor.

**31.** The communication device of claim 30, wherein the first processor is further configured to compare one or more of a value indicative of a distance, a value indicative of a battery power level, or a value indicative of signal quality.

**32.** The communication device of claim 30, wherein the RF modem is configured to sample the RF signals using a sample rate from 1 MHz to 12 Mhz.

**33.** The communication device of claim 32, wherein the digitized RF signals comprise I/Q data.

**34.** The communication device of claim 30, wherein the low bandwidth results comprise distance or time values.

**35.** The communication device of claim 30, wherein the first processor is configured transfer the low-bandwidth result over a first data path and transfer the digitized RF signals over a second data path.

**36.** The communication device of claim 35, wherein the first data path has a maximum data transfer rate that is lower than that of the second data path.

**37.** The communication device of claim 35, wherein the first processor is configured to transfer the digitized RF signals to a portion of a memory that is shared with the second processor.

**38.** The communication device of claim 37, wherein to transfer the digitized RF signals the first processor is configured to send a message to the second processor to cause the second processor to retrieve the digitized RF signals from the portion of the memory.

**39.** The communication device of claim 30, further comprising the second processor, wherein the

second processor is configured to execute an algorithm that uses the digitized RF signals to generate one or more location measurement values associated with the second communication device.

**40.** A system, comprising: a first processor and a second processor, a first data path and a second data path, each coupled between the first processor and the second processor; and a radio frequency (RF) modem coupled to the first processor, wherein the RF modem configured to: receive RF signals and generate digitized RF signals wherein the first processor configured to: operate in a first mode to generate low-bandwidth results using the digitized RF signals and transfer low-bandwidth results to the second processor via the first data path; compare one or more conditions to one or more threshold values; and transition, responsive to the comparing, from operating in the first mode to operating in a second mode; and while operating in the second mode, transfer the digitized RF signals to the second processor via the second data path, wherein the first data path has a maximum data transfer rate that is lower than that of the second data path.

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